Rearranging Formulae

Name _____ **Date** _____ No calculator Version 1

KEY WORDS formula · subject · rearrange · inverse operation · both sides

A Inverse operations ●○○ — ANSWER ON THE SHEET

Remember: An inverse operation undoes another operation. Subtracting 5 undoes adding 5, and dividing by 3 undoes multiplying by 3.

1. Write the operation that undoes each one.

a) $+5$ _____ **b)** $\times 3$ _____ **c)** -8 _____ **d)** $\div 4$ _____

e) $+a$ _____ **f)** $\times k$ _____ **g)** $-w$ _____ **h)** $\div t$ _____

2. Write the subject of each formula.

a) $A = lw$ _____ **b)** $d = st$ _____ **c)** $P = 4l$ _____ **d)** $y = x + 5$ _____

B Rearranging in one step ●●○ — WORK IN YOUR BOOK

Remember: The subject is the letter on its own. To change the subject, undo what has been done to the new letter, on both sides: $y = x + 5$ becomes $x = y - 5$.

3. Make the letter in square brackets the subject of each formula.

a) $y = x + 5 \quad [x]$ **b)** $y = x - 7 \quad [x]$ **c)** $y = 4x \quad [x]$

d) $y = \dfrac{x}{3} \quad [x]$ **e)** $A = lw \quad [l]$ **f)** $d = st \quad [t]$

g) $P = 4s \quad [s]$ **h)** $v = u + t \quad [u]$ **i)** $T = c - 4 \quad [c]$

j) $k = 10m \quad [m]$ **k)** $n = \dfrac{t}{7} \quad [t]$ **l)** $p = q + r \quad [q]$

C Rearranging in two steps ●●● — WORK IN YOUR BOOK

Watch out: Undo the operations in the reverse order. For $y = 2x + 3$, subtract 3 first, then divide by 2: $x = \dfrac{y - 3}{2}$.

4. Make the letter in square brackets the subject of each formula.

a) $y = 2x + 3 \quad [x]$ **b)** $y = 3x - 2 \quad [x]$ **c)** $C = 5n + 20 \quad [n]$

d) $y = 5x + 4 \quad [x]$ **e)** $k = 3m - 6 \quad [m]$ **f)** $y = \dfrac{x}{4} + 1 \quad [x]$

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D Using a rearranged formula ●●● WORK IN YOUR BOOK

5. The distance travelled at a steady speed is given by $d = st$.

a) Make t the subject of the formula.

b) Find t when $d = 120$ and $s = 40$.

6. The area of a rectangle is given by $A = lw$.

a) Make w the subject of the formula.

b) Find w when $A = 48$ and $l = 8$.

7. The perimeter of a square is given by $P = 4s$.

a) Make s the subject of the formula.

b) Find s when $P = 36$.

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8. The cost in dollars of hiring a hall for n hours is given by $C = 5n + 20$.

a) Make n the subject of the formula.

b) Find n when $C = 65$.

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Rearranging Formulae

Name _____ **Date** _____ No calculator **Version 2**

KEY WORDS formula · subject · rearrange · inverse operation · both sides

A Inverse operations ●○○ — ANSWER ON THE SHEET

Remember: An inverse operation undoes another operation. Subtracting 5 undoes adding 5, and dividing by 3 undoes multiplying by 3.

1. Write the operation that undoes each one.

a) $+7$ _____ **b)** $\times 2$ _____ **c)** -6 _____ **d)** $\div 5$ _____

e) $+b$ _____ **f)** $\times m$ _____ **g)** $-h$ _____ **h)** $\div s$ _____

2. Write the subject of each formula.

a) $A = bh$ _____ **b)** $v = u + t$ _____ **c)** $P = 3s$ _____ **d)** $y = x - 2$ _____

B Rearranging in one step ●●○ — WORK IN YOUR BOOK

Remember: The subject is the letter on its own. To change the subject, undo what has been done to the new letter, on both sides: $y = x + 5$ becomes $x = y - 5$.

3. Make the letter in square brackets the subject of each formula.

a) $y = x + 8 \quad [x]$ **b)** $y = x - 3 \quad [x]$ **c)** $y = 6x \quad [x]$

d) $y = \dfrac{x}{5} \quad [x]$ **e)** $A = bh \quad [b]$ **f)** $d = st \quad [s]$

g) $P = 3s \quad [s]$ **h)** $v = u + t \quad [t]$ **i)** $T = c - 9 \quad [c]$

j) $k = 100m \quad [m]$ **k)** $n = \dfrac{t}{4} \quad [t]$ **l)** $a = b + c \quad [b]$

C Rearranging in two steps ●●● — WORK IN YOUR BOOK

Watch out: Undo the operations in the reverse order. For $y = 2x + 3$, subtract 3 first, then divide by 2: $x = \dfrac{y - 3}{2}$.

4. Make the letter in square brackets the subject of each formula.

a) $y = 4x + 1 \quad [x]$ **b)** $y = 2x - 5 \quad [x]$ **c)** $C = 8n + 15 \quad [n]$

d) $y = 6x + 5 \quad [x]$ **e)** $k = 4m - 8 \quad [m]$ **f)** $y = \dfrac{x}{3} + 2 \quad [x]$

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D Using a rearranged formula ●●● WORK IN YOUR BOOK

5. The distance travelled at a steady speed is given by $d = st$.

a) Make s the subject of the formula.

b) Find s when $d = 150$ and $t = 3$.

6. The area of a rectangle is given by $A = lw$.

a) Make l the subject of the formula.

b) Find l when $A = 54$ and $w = 6$.

7. The perimeter of a square is given by $P = 4s$.

a) Make s the subject of the formula.

b) Find s when $P = 52$.

★ Extension FOR EARLY FINISHERS

8. The cost in dollars of hiring a boat for n hours is given by $C = 8n + 15$.

a) Make n the subject of the formula.

b) Find n when $C = 71$.

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